Publication review and revision record

Neural means and kernel corrections for operator learning
Yitzchak Shmalo | 5 September 2026

Original manuscript: NO. My vote is NO; the twenty separately spawned reviewing agents returned **0 YES and 20 NO** on version e unchanged. All twenty individual reports are included in the source ZIP.

Revised manuscript: YES for submission as an empirical research paper. The revision retains the supported measurements and limits its claims to the tested estimators, metrics, and predictor pools. This is a submission recommendation, not a prediction of journal acceptance.

The votes are separate AI-agent assessments with different audit emphases. They are not external journal reviews or a statistical sample of independent experts; shared parent context and model limitations can affect them. The twenty votes apply to the original. Follow-up reviews of selected revisions were targeted checks, not a new twenty-person vote.

What the checks established

The public source snapshot 50c05ff1cce32d20df446c5c57c1c20027df5875 contains a main PDF whose extracted text matches the supplied 39-page version exactly. The repository macro audit was rerun and found zero discrepancies with its summary artifacts. Its fifteen algebraic checks passed. Reviewers independently recomputed the sixty-member empirical RMS lower bound as 4.813611379%, versus the stored hindsight optimum 4.876132938%. These are summary/matrix-level checks; the neural training campaigns were not rerun.

The OCO-2 original thirty-run summaries support the headline means and the weak-CO2 one-seed radiance exception. The coordinate combination slightly trails its flat feature head in reduced error on O2 and weak CO2, so the claim of beating every constituent on both metrics was removed.

Caltech access

The user authorized the Caltech/DGX machine for simulations and source retrieval. A normal SSH attempt failed before authentication with “Network is unreachable.” No remote files were read and no simulations were started there. The matching LaTeX came from the public repository. This package claims no results from inaccessible corrected-centering or wider-grid runs.

Twenty publication votes

ID	Vote	Principal reason for NO on the original
01	NO	Theorem endpoints, margin tie case, and mean-versus-RMS claims.
02	NO	Recorded estimator differs from strict cross-fitting; metric and inference claims.
03	NO	Novelty is empirical; representation claims and artifact availability need precision.
04	NO	Equicorrelation diagnostic mislabeled as rigorous floor; in-sample versus transfer.
05	NO	RKHS theorem scope, correction-label mismatch, and conformal qualifications.
06	NO	OCO-2 metric attribution and false dominance over every component.
07	NO	Only the kernel channel is foldwise; pooled centering and test-access disclosure.
08	NO	Boundary counterexamples, arbitrary-weight bracket overclaim, and missing radius.
09	NO	Unsupported statistical equivalence and future-pool extrapolation.
10	NO	Useful empirical contribution, but categorical kernel and floor claims too strong.
11	NO	Ridge is not the noiseless minimax interpolant; mixed-label bound needed.
12	NO	Augmented ERM is not symmetrized ERM; binary decision needed at a tie.
13	NO	Direct-target versus residual heads, metrics, and limited ridge-control conclusion.
14	NO	Formal endpoint conditions and explicit result-to-proof references.
15	NO	Numerical macros check out; estimator and comparison descriptions need repair.
16	NO	Conformal no-ties assumptions, exact reference law, and infinite quantile endpoint.
17	NO	Empirical sixty-member bound correct; finite-pool and objective scope overstated.
18	NO	Model label mlpR misidentified; table claims and standard-error arithmetic.
19	NO	Unsupported anisotropic rates and exact mathematical counterexamples.
20	NO	Claim-table contradictions, recorded centering, and incomplete proof pointers.

Each full report appears as reviews/review01.md through reviews/review20.md in the ZIP. The reports preserve the reviewers' original judgments; later repairs are recorded separately.

Substantive changes

Estimator and protocol. Specified the procedure that produced the reported mechanics scores: foldwise kernel weights with pooled target centering, in-sample neural training predictions, and a changed refiner channel at deployment. Retained the old measurements under that definition and made no claim that later fold-local code reproduced them. Repeated test-block access is disclosed.

Kernel error bound. Added and proved the explicit correction-label mismatch term. Corrected the missing ball-radius factor in the exact worst-case statement, the scaled sharpness-proof row, and the distinction between interpolation minimaxity and positive-ridge error. Fitted norms are not treated as certified norms of the query-time residual.

Ensemble theory. Separated mean relative error from RMS error, exact empirical floors from equicorrelation diagnostics, and optimized global convex mixtures from per-pixel or corrected estimators. Preserved the verified sixty-member floor and corrected the six-member floor values. Removed the supposed prospective validation of a same-validation dispersion ratio.

Mathematical repairs. Excluded perfect anticorrelation from strict derivative claims; treated zero-risk and zero-floor denominators; changed the margin rule to a binary decision with a stated tie convention. Replaced the unsupported anisotropic-rate argument with a finite-design Lipschitz rank-truncation proposition and full proof, independently checked by reviewer 19.

Coverage. Stated the separate hypotheses for marginal conformal coverage, the no-ties upper bound, and the i.i.d. continuous-score Beta reference law. Added the infinite-quantile endpoint. The retrospective coverage experiment is not presented as proof of fresh-population independence.

OCO-2. Wrote reconstruction and scoring formulas explicitly, identified the metrics as retained-representation rescoring criteria, corrected selector tradeoffs and rounding language, filled the ARD radiance cell, and restricted conclusions about ridge readouts and matched tuning budgets.

Presentation and sources. Rewrote the abstract, introduction, discussion, and substantial experimental prose in direct academic language. Added the supplement to the GitHub footnote and a specific supplement proof pointer beside every main formal result. Preserved funding and competing-interest information and retained an AI-use declaration. Added bibliography identifiers and made source/archive availability precise.

Scope after revision

The paper supports a benchmark study of established neural/kernel constructions and a rigorous analysis of the measured predictor pool. It does not establish a universal benchmark noise floor, improved end-to-end OCO-2 retrieval, statistical equivalence to unavailable baseline predictions, or new results from simulations that were not run. The full derivations and additional measurements remain in the accompanying supplement.

Overleaf package

Import the ZIP as a new project, choose pdfLaTeX, and select main.tex. The included latexmk configuration builds the companion supplement and resolves references in both directions. To view the supplement as the editor output, select supplement.tex. The full source, figures, bibliography, twenty reviews, targeted follow-up audits, and verification logs are included.